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Structural Studies in Lithium Insertion into SnO-B_2O_3 Glasses and Their Applications for All-Solid-State Batteries

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The local structure of electrochemically lithium-inserted SnO-B_2O_3 glasses was investigated by several spectroscopic techniques to clarify a lithium insertion mechanism into the glasses. $50SnO \cdot 50B_2O_3$ (mol %) glass showed two plateaus around 1.5 and 0.5 V (*vs.* Li^+/Li) on the lithium insertion process and exhibited a high capacity of 1240 mAh g^{-1} in the case of using a conventional liquid electrolyte. On the first plateau (1.5 V *vs.* Li^+/Li), metallic Sn with small domains was formed and the coordination environment at boron in the glass network was not changed. On the second plateau (0.5 V *vs.* Li^+/Li), the borate glass network was rearranged by a transformation from tetrahedral BO_4 to trianglar BO_3 boron units, which provides an additional free space compensating an increase in volume followed by a formation of Li-Sn alloy domains. Hence, the larger the fraction of tetrahedral BO_4 unit is in the SnO-B_2O_3 glasses, the higher the charge-discharge capacities are. The SnO-B_2O_3 glasses are applicable to all-solid-state lithium rechargeable batteries as anode materials with high capacity.
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Tin oxide-based glasses and crystals have been studied extensively as anode materials for lithium secondary batteries since the first report by Idota *et al.*[1-9] These anode materials show high specific capacities twice that of carbon materials, which are commercially used as anode materials for lithium-ion secondary batteries.

Recently, we reported that the binary system SnO-B_2O_3 has a wide glass-forming region with $0 \leqslant$ SnO (mol %) $\leqslant 75$, and these glasses work as an anode material with high capacity using a conventional cell with liquid electrolytes.[10] Charge-discharge capacities depend on the glass composition and are maximized at the composition of 50 mol % SnO. ^{11}B NMR measurements have revealed that four-coordinated boron in the glass network is maximized at the composition of 50 mol % SnO,[11] suggesting that there is a close relationship between electrochemical capacity and local glass structure.

To clarify the origin of the relationship, an investigation of structural changes during electrochemical lithium insertion to the SnO-B_2O_3 binary glasses is important. Structural studies on electrochemical reaction of lithium with SnO-based glasses by several spectroscopic techniques have been reported for multi-oxide systems such as SnO-B_2O_3-P_2O_5-Al_2O_3.[5,6,8] Those studies for the simple binary system SnO-B_2O_3, however, have not been examined in detail.

Nowadays, developing all-solid-state lithium rechargeable batteries with high reliability and high safety is desirable to replace the current lithium-ion batteries using liquid electrolytes, which have inherently irresolvable problems of leakage and inflammability. All-solid-state electrochemical cells consisting of Li-In alloy or In (anode), a superionic oxysulfide glass (solid electrolyte), and $LiCoO_2$ (cathode) have been reported to work as lithium secondary batteries and exhibit excellent cycling properties.[12-15] SnO-based glasses with high capacity are expected to be applied for anode materials of all-solid-state lithium secondary batteries

In this study the local structure of the SnO-B_2O_3 binary glasses to which lithium ions are electrochemically inserted from a conventional liquid electrolyte is analyzed in detail by several spectroscopic techniques. Based on the obtained structural data, we have proposed an idea regarding the lithium insertion mechanism to the SnO-based glasses from the viewpoint of a rearrangement of glass network. Effects of glass structure on charge-discharge capacities of these glasses have also been discussed. An all-solid-state cell of the SnO-B_2O_3 glass (anode)/Li_2S-SiS_2-Li_3BO_3 glass (solid electrolyte)/$LiCoO_2$ (cathode) is practically assembled. From its charge-discharge performances we have considered the potential of the SnO-B_2O_3 glasses for application to all-solid-state lithium rechargeable batteries.

Experimental

Glass preparation.—A mixture of SnO (Furuuchi Chem., 99.9%) and B_2O_3 (Koujundo Chem., 99.9%) was heated in a carbon crucible at 900-1200°C for 30 min in a dry N_2-filled glove box. A molten sample was cooled by a conventional press-quenching method (the cooling rate is 10^2 to 10^3 K/s). The obtained materials were transparent glasses, which showed a halo pattern in X-ray diffraction (XRD) measurements and glass transition phenomena in differential scanning calorimetry (DSC) measurements.

Cell assembly.—The obtained glasses were ground into fine powder by a planetary ballmill apparatus (Frich Pulversette 7). Electrode materials were prepared by mixing the glass powder (70 wt %), acetylene black (20 wt %), and polyvinylidene fluoride (PVDF, 10 wt %) with an agate mortar in the dry Ar-filled glove box. Slurries of the mixture were painted on a Ni mesh (1×1 cm) as a current collector and dried for 2 h at 150°C. The prepared electrode was pressed between stainless steel plates under 2900 kg/cm^2. Electrochemical measurements were performed in a simple three-electrode cell with the obtained electrode as a working electrode and lithium sheet as both counter and reference electrodes. A mixture of ethylene carbonate (EC) and diethyl carbonate (DEC) with 1:1 volumetric ratio containing 1 mol dm^{-3} (1 M) $LiPF_6$ was used as an electrolyte.

An all-solid-state electrochemical cell was assembled as follows. The $95(0.6Li_2S \cdot 0.4SiS_2) \cdot 5Li_3BO_3$ (mol %) superionic glass with high conductivity of 10^{-3} S cm^{-1} at room temperature was used as a solid electrolyte.[13,16,17] The glass was obtained by melt-quenching and then was ground into fine powder by ballmilling. A positive electrode was prepared by mixing $LiCoO_2$ (38 wt %), the glass electrolyte (57 wt %), and acetylene black (5 wt %) powders. A negative electrode was obtained by mixing the $50SnO \cdot 50B_2O_3$ (mol %) glass (42 wt %), the glass electrolyte (53 wt %), and acetylene black (5 wt %) powders. The positive electrode powder (100 mg), the glass electrolyte powder (80 mg), and the negative electrode powder (19 mg) were put together in an insulator tube (polycarbon-

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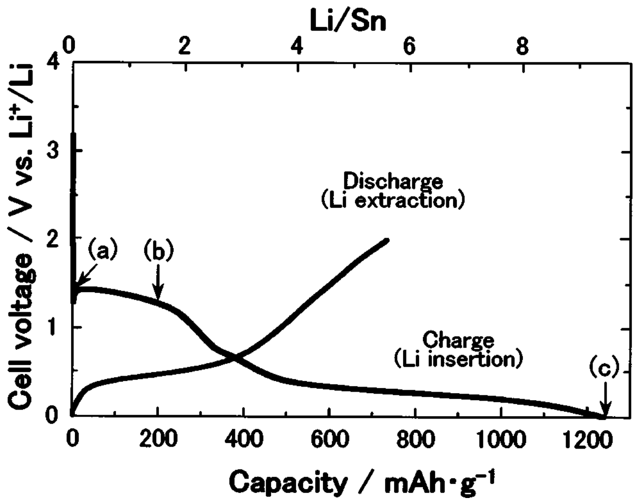

Figure 1. First charge-discharge profile for a cell of $Li/50SnO \cdot 50B_2O_3$ (mol %) glass. A mixture of 1 M $LiPF_6$/EC + DEC (1:1) is used as a liquid electrolyte. The measurement is performed at room temperature under a constant current density of 1.0 mA cm^{-2} between fixed voltage limits of 0.0 and 2.0 V.

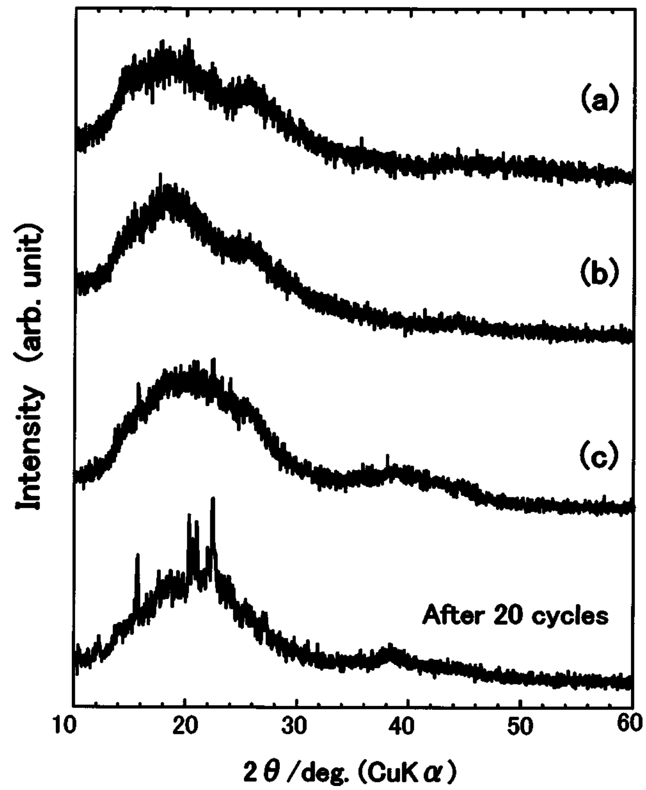

Figure 2. XRD patterns for $50SnO \cdot 50B_2O_3$ (mol %) glasses at various levels of charge (a-c) as shown on the voltage profile in Fig. 1: (a) Initial state, (b) 200, and (c) 1240 mAh g^{-1} (the cell voltage reaches 0 V). The XRD result of the glass after 20 charge-discharge cycles is also shown in this figure.

ate, ϕ = 10 mm) and then were pressed under 3700 kg/cm^2 to obtain an all-solid-state cell.

Structural analysis.—After lithium ions were electrochemically inserted in the three-electrode cell using the liquid electrolyte mentioned above, the working electrode material including the $50SnO \cdot 50B_2O_3$ glass was scraped from Ni mesh. The obtained powder material was dried and then used for the following spectroscopic measurements.

XRD measurements were carried out using a Shimadzu XRD-6000 diffractometer employing a Cu Kα source. Powder samples were glued on a glass plate and covered with polyimide thin film in an Ar atmosphere to avoid the attack of water and oxygen in air.

^{11}B and ^{7}Li magic-angle spinning (MAS)-NMR measurements were performed on a Varian Unity Inova 300 NMR spectrometer. The powder samples were packed into a zirconia spinner with a sealant in a dry Ar-filled glove box. A pulse sequence with a single $\pi/8$ pulse was used to perform quantitative analysis on the spectra of quadrupolar nuclei such as ^{11}B.[18] In the ^{11}B nucleus, a width of a $\pi/8$ pulse length of 0.7 μs, a recycle pulse delay of 2.0 s, and a spinning speed of 5 kHz were used. BPO_4 crystal powder was used as a standard material for ^{11}B NMR measurements. In the ^{7}Li nucleus, a width of a $\pi/2$ pulse length of 5 μs, a recycle pulse delay of 10 s, a spinning speed of 5 kHz, and a standard material of 1 M LiCl solution were used.

^{119}Sn Mössbauer spectra were measured with a constant acceleration mode Mössbauer spectrometer using a $CaSnO_3$ source at room temperature. The spectra were obtained by Wissel spectrometer with a NaI (TI) scintillation counter and recorded on a Seiko 7700 multichannel analyzer.

Results

Initial charge and discharge profile.—Figures 1 shows the first charge-discharge profile for a cell of $Li/50SnO \cdot 50B_2O_3$ (mol %) glass. A mixture of 1 M $LiPF_6$/EC + DEC (1:1) was used as a liquid electrolyte. The measurement is performed at room temperature under a constant current density of 1.0 mA cm^{-2} between fixed voltage limits of 0.0 and 2.0 V. In this figure, a charge means an insertion of lithium to a working electrode ($50SnO \cdot 50B_2O_3$ glass), while a discharge means an extraction of lithium from the working electrode. The abscissa shows the charge-discharge capacities per unit weight of glass. Two plateaus around 1.5 and 0.5 V are observed on the first charge curve, while only one plateau around 0.5 V is observed on the discharge curve. The charge-discharge profiles the $50SnO \cdot 50B_2O_3$ glass are similar to those of other SnO-based multi-oxide glasses[1-9] obtained so far. The charge capacity of the glass exhibits a high value of 1240 mAh g^{-1}, whereas the discharge capacity shows 733 mAh g^{-1}. Although a large irreversible capacity is present at the first cycle, the glass works as an anode material with high capacity for lithium secondary batteries using a conventional liquid electrolyte.

Structural change during lithium insertion.—To elucidate the lithium insertion mechanism to the $50SnO \cdot 50B_2O_3$ glass we have investigated structural changes of the glass during the first charge.

Figure 2 shows the XRD patterns for $50SnO \cdot 50B_2O_3$ glasses at various levels of charge as shown on the voltage profile in Fig. 1. The diffraction pattern of the glass after 20 charge-discharge cycles is also shown in this figure. Halo patterns are basically observed in all samples. No obvious diffraction peaks appear in sample c but intensities at around $2\theta =$ 22 and 38° become stronger rather than those in samples a and b. Diffraction peaks at those positions are clearly observed in the sample after 20 charge-discharge cycles. Courtney and Dahn have reported that broad peaks at approximately 2θ = 22 and 38° are observed on the process of lithium insertion to SnO-based glasses and these peaks are due to Li-rich alloy phases such as Li_5Sn_2, $Li_{13}Sn_5$, Li_7Sn_2, and $Li_{22}Sn_5$ with poorly developed long-range order.[2] Figure 2 thus suggests that those Li-Sn alloy phases are formed in sample c.

Figure 3 shows the ^{11}B MAS-NMR spectra for the pristine $50SnO \cdot 50B_2O_3$ glass and the glasses charged to 200 and 1240 mAh g^{-1}. The sharp peak at 0 ppm is due to four-coordinated boron BO_4

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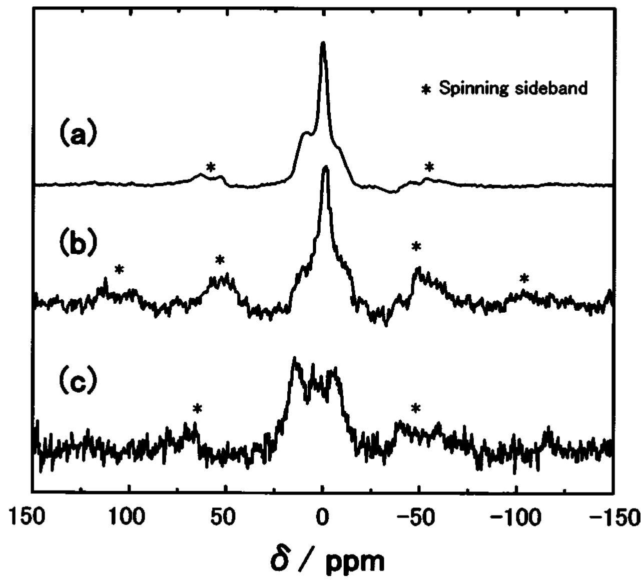

Figure 3. ^{11}B MAS-NMR spectra for (a) pristine $50SnO \cdot 50B_2O_3$ glass and glasses charged to (b) 200 and (c) 1240 mAh g^{-1}.

and the broad peak from −20 to +20 ppm is due to three-coordinated boron BO_3.[11,19] Both four- and three-coordinated borons are present in $50SnO \cdot 50B_2O_3$ glass. Little change between Fig. 3a and b is observed in the peak position and shape of spectrum. Spectrum 3c is drastically changed from spectra a and b; the intensity of the sharp peak at 0 ppm decreases in spectrum c. Figure 3 reveals that the borate glass network in sample b is almost the same as that in the pristine glass, while the glass network has been rearranged in sample c due to the decrease in four-coordinated boron.

Figure 4 shows the ^{7}Li MAS-NMR spectra for $50SnO \cdot 50B_2O_3$ glasses charged to 200 and 1240 mAh g^{-1}. Only one peak at 0 ppm is observed in sample b, while a new peak at around 5 ppm in addition to the peak at 0 ppm is observed in sample c. The peak at 0

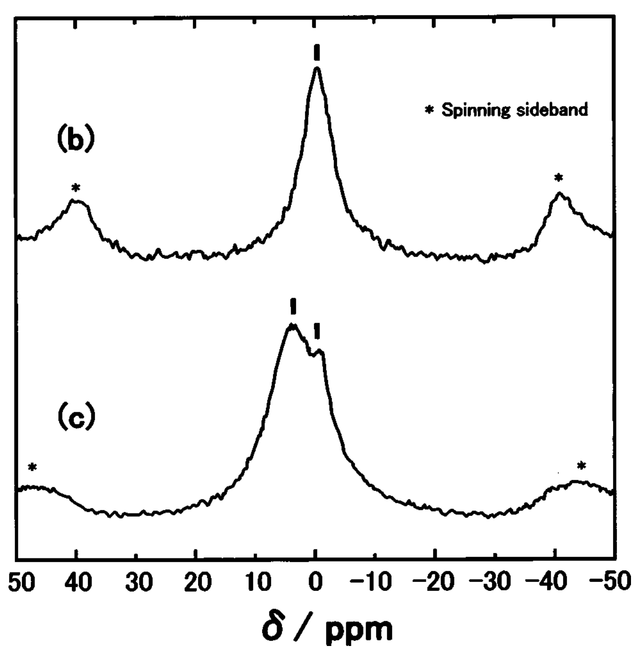

Figure 4. ^{7}Li MAS-NMR spectra for $50SnO \cdot 50B_2O_3$ glasses charged to (b) 200 and (c) 1240 mAh g^{-1}.

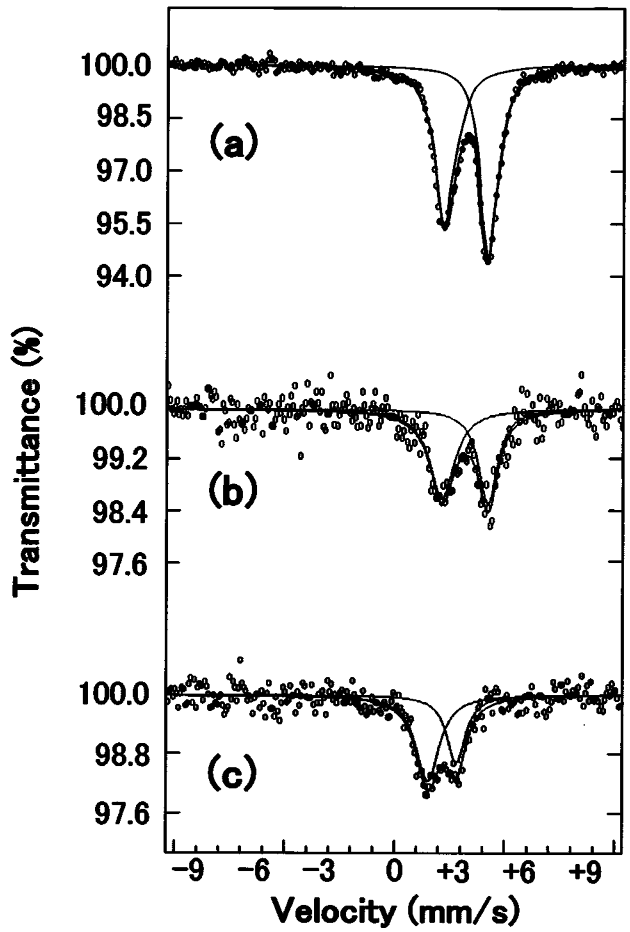

Figure 5. ^{119}Sn Mössbauer spectra for (a) pristine $50SnO \cdot 50B_2O_3$ glass and glasses charged to (b) 200 and (c) 1240 mAh g^{-1}.

ppm suggests that lithium is present in an ionic state like in LiCl ionic crystal (a standard material). Goward *et al.*[8] have reported that Li-Sn alloy phases electrochemically precipitated in the SnO-based glasses show peaks between 1 and 12 ppm in ^{6,7}Li NMR spectra where the resonance is shifted to the higher frequency side with an increase in the inserted lithium concentration. They have also pointed out that the Li-Sn domains are limited in size in $SnO\text{-}Al_2O_3\text{-}B_2O_3\text{-}P_2O_5$ glass rather than in SnO crystal because the oxide glass matrix prevents the aggregation of those Li-Sn particles. Hence, Fig. 4 suggests that sample b includes only ionic lithium and sample c contains not only ionic lithium but also Li-Sn alloy phases with small domains.

Figure 5 shows the ^{119}Sn Mössbauer spectra for pristine $50SnO \cdot 50B_2O_3$ glass and the glasses charged to 200 and 1240 mAh g^{-1}. The spectrum of sample a is doublet and its shape and peak positions are similar to those of SnO [Sn(II)] crystal, suggesting that the oxidation state of Sn in pristine glass is only Sn(II).[11] The peaks are broadened but their positions are not changed in sample b. The peak positions shift slightly to the lower velocity direction in sample c. Courtney *et al.*[6] have reported that a broad peak due to metallic Sn is observed in the Mössbauer spectrum of the $SnO\text{-}B_2O_3\text{-}P_2O_5$ glass at the charging level just after a first plateau. The spectrum of sample b suggests that the sample still includes Sn(II) which shows sharp doublet peaks and thus there is no obvious evidence of metallic Sn, which shows only a weak-broad peak. This would be because the first plateau is not completely finished at the charging level of sample b. Compared with the shape and position of Mössbauer spectra of the lithium-inserted glasses,[6] the spectrum of sample c is

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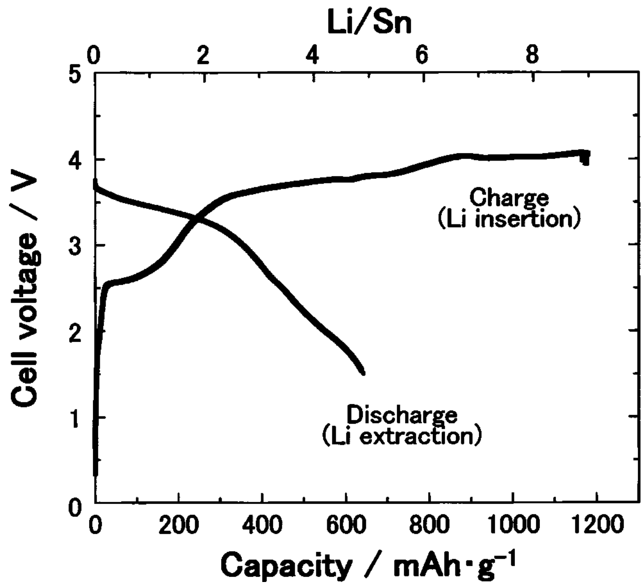

Figure 6. First charge-discharge profile for an all-solid-state cell of $50SnO \cdot 50B_2O_3$ glass/$LiCoO_2$. A superionic glass in the system Li_2S-SiS_2-Li_3BO_3 is used as a solid electrolyte. The measurement is performed at room temperature under a constant current density of 64 $\mu A\ cm^{-2}$ between fixed voltage limits of 1.5 and 4.1 V. The abscissa denotes capacities per unit weight of the anode glass.

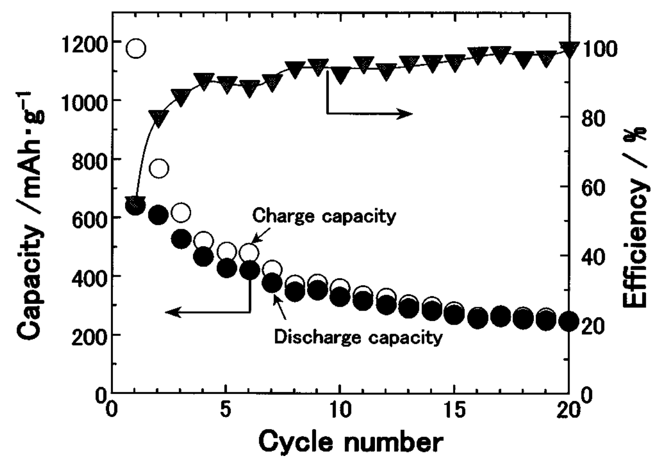

Figure 7. Cycle performance on charge-discharge capacities and its efficiency for an all-solid-state cell of $50SnO \cdot 50B_2O_3$ glass/$LiCoO_2$ under the same condition as shown in Fig. 6.

assigned to Li-rich Li-Sn alloy phases such as Li_5Sn_2, $Li_{13}Sn_5$, Li_7Sn_2, and $Li_{22}Sn_5$.

Electrochemical property of an all-solid-state cell.—Figure 6 shows the first charge-discharge profile for an all-solid-state cell of $50SnO \cdot 50B_2O_3$ glass/$LiCoO_2$. The $95(0.6Li_2S \cdot 0.4SiS_2) \cdot 5Li_3BO_3$ (mol %) superionic glass with high conductivity of 10^{-3} S cm^{-1} at room temperature is used as a solid electrolyte.[13,16,17] The measurement is performed at room temperature under a constant current density of 64 $\mu A\ cm^{-2}$ between fixed voltage limits of 1.5 and 4.1 V. The abscissa denotes capacities per unit weight of the anode glass. Two plateaus around 2.5 and 3.5 V are observed on charge and then one plateau around 3.5 V is observed on discharge. The two plateaus observed on charge correspond to those of 1.5 and 0.5 V observed in the cell using a liquid electrolyte as shown in Fig. 1. Total charge and discharge capacities are, respectively, 1180 and 640 mAh g^{-1}.

Figure 7 shows the cycling performance on charge-discharge capacities and its efficiency for the all-solid-state cell. Both charge and discharge capacities gradually decrease with cycling while the charge-discharge efficiency increases. After 20 cycles the efficiency reaches 100% and the cell maintains capacities over 200 mAh g^{-1}.

Discussion

Relationship between capacity and structure.—Figure 8 shows the composition dependence of the first charge-discharge capacities of the SnO-B_2O_3 glasses using a mixture of 1 M $LiPF_6$/EC + DEC (1:1) as a liquid electrolyte. These capacities are obtained by measurements at room temperature under a constant current density of 1.0 mA cm^{-2} between fixed voltage limits of 0.0 and 2.0 V. The fraction of four-coordinated boron (N_4) in the glasses, which was obtained in our previous paper,[11] is also shown in this figure.

The discharge capacity exhibits almost the same composition behavior as the charge capacity. These capacities increase up to the composition with 50 mol % SnO, then decrease up to 67 mol % SnO, and increase again up to 75 mol % SnO. The maximum charge and discharge capacities are, respectively, 1240 and 733 mAh g^{-1} at the composition with 50 mol % SnO. The value of N_4 increases and then decreases with an increase in SnO content. The maximum N_4 of 0.28 is obtained at the composition with 50 mol % SnO. The composition at which charge-discharge capacities are maximized is the same as the composition at which N_4 is maximized.

Generally, capacities are expected to increase with an increase in SnO content because the amount of Sn, which works as an active site for lithium insertion, increases. For example, charge-discharge capacities increased monotonically with an increase in SnO content in the SnO-BPO_4 system.[10] In the SnO-B_2O_3 system, however, charge-discharge capacities have a maximum at the composition

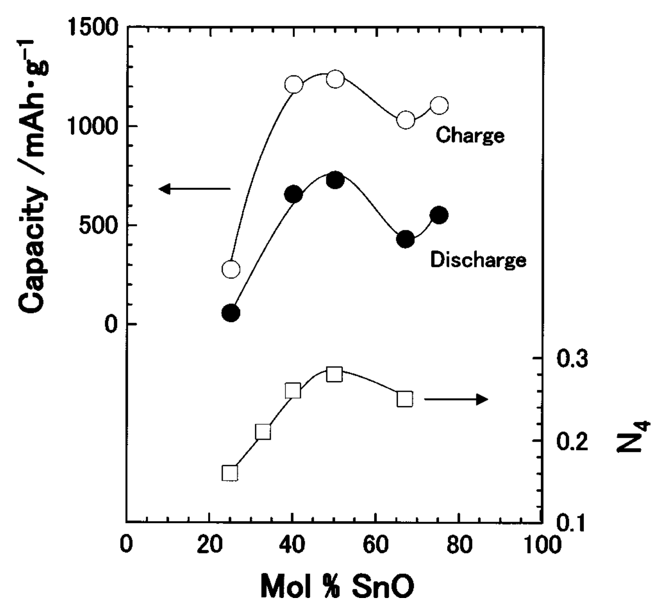

Figure 8. Composition dependence of the first charge and discharge capacities of SnO-B_2O_3 glasses. These capacities are obtained by measurements under the same condition as shown in Fig. 1. Fraction of four-coordinated boron (N_4) in the glasses, which was obtained in our previous paper,[11] is also shown in this figure.

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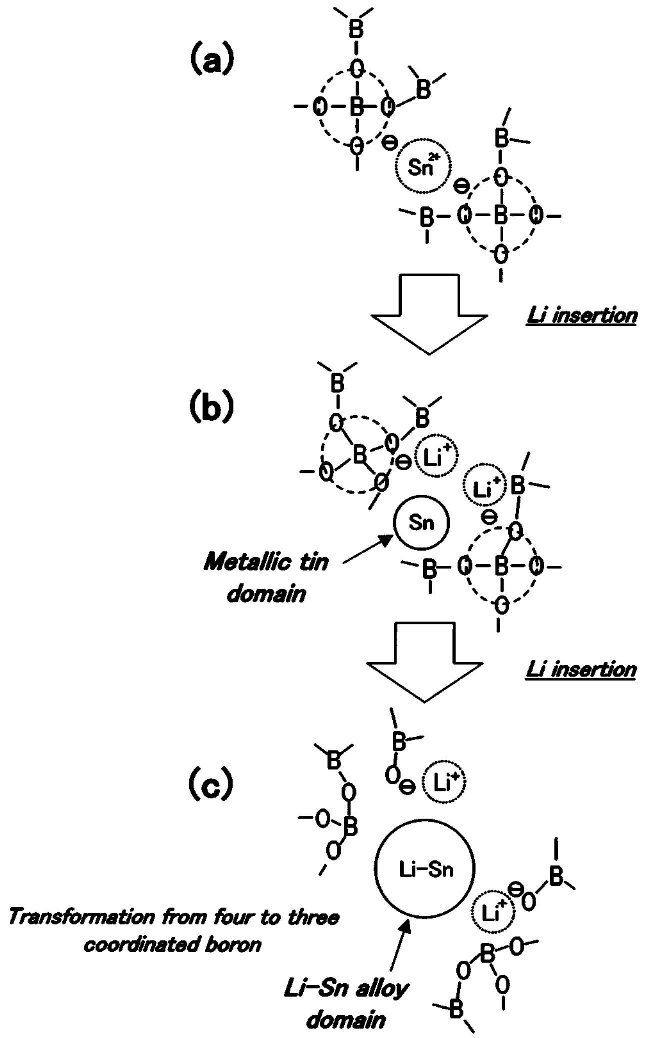


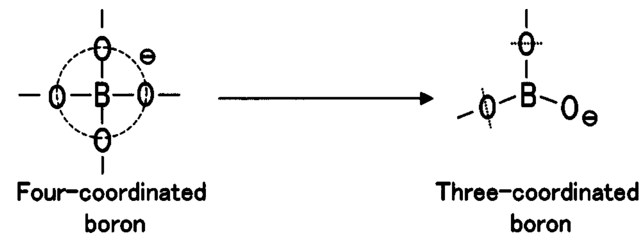
Figure 9. Electrochemical lithium insertion model of $50SnO \cdot 50B_2O_3$ glass.

with 50 mol % SnO, suggesting that the capacities are affected by not only SnO content but also glass structure (particularly, fraction of four-coordinated boron) in the system. The reason for showing large capacities at the composition with a large fraction of four-coordinated boron is discussed in the following section from the viewpoint of glass structural changes during lithium insertion.

Mechanism of lithium insertion.—Based on glass structural data obtained by several spectroscopic measurements, the lithium insertion mechanism has been proposed. Figure 9 shows a model of the electrochemical lithium insertion in $50SnO \cdot 50B_2O_3$ glass. In the pristine glass before lithium insertion, the glass network consists of both three-coordinated boron (the trianglar $BO_{3/2}$ unit) and four-coordinated boron [the tetrahedoral $(BO_{4/2})^-$ unit]. Tin is present as Sn^{2+}, which compensates negative charges of two $(BO_{4/2})^-$ units.

In Fig. 9b after lithium insertion up to the charge capacity of 200 mAh g^{-1}, the borate glass network is the same as that in pristine glass. ^{7}Li NMR spectrum suggests that the inserted lithium is present as an ionic state (Li^+). In this study, there is no obvious evidence of metallic Sn from XRD and ^{119}Sn Mössbauer measurements. However, metallic Sn should be formed in this stage because Li^+ would compensate a negative charge of the $(BO_{4/2})^-$ unit instead of Sn^{2+}.

In Fig. 9c after lithium insertion up to the charge capacity of 1240 mAh g^{-1}, four-coordinated boron in the glass network is reduced, suggesting that a transformation from four-coordinated boron [the tetrahedral $(BO_{4/2})^-$ unit] to three-coordinated boron [the trianglar $(BO_{2/2})O^-$ unit] shown in Scheme 1



Scheme 1. Transformation from four-coordinated boron to three-coordinated boron.

occurs. It is revealed from XRD, ^{7}Li NMR, and ^{119}Sn Mössbauer measurements that Li-rich Li-Sn alloy phases with small domains are present in this stage.

Here, we discuss in detail the glass network change between Fig. 9b and c, which corresponds to the electrochemical reaction of the plateau of 0.5 V on the charge process (Fig. 1). As shown in Scheme 1, the rearrangement of glass network by the transformation from the tetrahedral $(BO_{4/2})^-$ unit to the trianglar $(BO_{2/2})O^-$ unit occurs with lithium insertion in this stage.

The individual volumes of those borate units have been obtained from the density data in the alkali borate glasses.[20-22] In the Li_2O-B_2O_3 system, the relative volume of the tetrahedral $(BO_{4/2})^-$ unit compared with the $BO_{3/2}$ unit in pure B_2O_3 glass is 0.94, which is smaller than the value of 1.28 of the trianglar $(BO_{2/2})O^-$ unit. Those relative volumes, however, include empty (free) space in addition to the space originally occupied by the borate unit. To consider a structural arrangement of a glass, the concept of "packing fraction" proposed by Feller *et al.*[20-22] is appropriate and useful. The packing fraction is defined as a ratio of the volume calculated from ionic radii to the volume found from the measured density including free space. The calculated packing fraction of the $(BO_{4/2})^-$ and $(BO_{2/2})O^-$ units are, respectively, 0.65 and 0.39;[22] the tetrahedral $(BO_{4/2})^-$ unit shows a noticeably higher packing fraction than the planar trianglar $(BO_{2/2})O^-$ unit. These results suggest that the transformation from the $(BO_{4/2})^-$ unit to the $(BO_{2/2})O^-$ unit provides an additional free space in the glass network.

According to previous studies,[2,8] the size of Li-Sn alloy particles gradually increases followed by lithium insertion in the SnO-based glasses and the oxide glassy matrix plays an important role in preventing the aggregation of the Li-Sn alloy particles. In the SnO-B_2O_3 glasses including four-coordinated boron, the additional free space generated by the transformation from the tetrahedral unit to the trianglar unit can compensate the volume-increase of the alloy domains, suggesting that further amounts of lithium could be easily inserted and Sn active sites would be much more effectively utilized in those glasses. Hence, the unusually large capacity is obtained at the composition of relatively small ratio of SnO ($50SnO \cdot 50B_2O_3$), where the fraction of four-coordinated boron is maximized (Fig. 8).

Application to all-solid-state rechargeable batteries.—We have compared the electrochemical properties of the solid electrolyte system with those of the liquid electrolyte system. The all-solid-state cell of $50SnO \cdot 50B_2O_3$ glass/$LiCoO_2$ exhibits high charge and discharge capacities of 1180 and 640 mAh g^{-1} at the first cycle (Fig. 6) under a constant current density of 64 μA cm^{-2} between the voltage limits of 1.5 and 4.1 V. These capacity values are comparable to the charge and discharge capacities of 1240 and 733 mAh g^{-1} in the liquid electrolyte cell of Li/$50SnO \cdot 50B_2O_3$ glass (Fig. 1) obtained under a constant current density of 1.0 mA cm^{-2} between the voltage limits of 0.0 and 2.0 V. The capacities are reduced to about 200 mAh g^{-1} and charge-discharge efficiency increases up to 100% after

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20 cycles in the solid electrolyte system and the cycling behavior is almost the same as in the liquid electrolyte system described in our previous paper.[10]

The problem of decreasing in capacity with cycling can be overcome by selecting appropriate voltage ranges. In the liquid electrolyte cell of Li/the $50SnO \cdot 50B_2O_3$ glass, the cycling performance is considerably improved by reducing the upper cutoff voltage from 2.0 to 0.8 V.[10] The formation of metallic Sn and its aggregation would occur in the voltage range over 0.8 V at the first discharge (lithium extraction).[3] The large volume change due to the aggregated Sn particles during charge-discharge cycling probably causes the degradation of capacity. Optimizing voltage limits would also improve the cycling performance of all-solid-state cells.

The large irreversible capacity was observed during first cycle in both liquid and solid electrolyte systems. This capacity loss mainly corresponds to the capacity of the first plateau on the charging process (the plateau of 1.5 V in Fig. 1), where metallic tin is formed and lithium ions (Li^+) compensate negative charges of borate glass network. These lithium ions are strongly trapped by anionic oxide network and, therefore, it is difficult to extract the lithium ions on the discharge process electrochemically.

Although there are a few unsolved issues of improving cycle performance, the $SnO\text{-}B_2O_3$ glasses are one of the most promising anode materials for all-solid-state lithium rechargeable batteries.

Conclusions

We prepared $SnO\text{-}B_2O_3$ glasses by melt-quenching and investigated their electrochemical properties in not only a conventional liquid electrolyte cell but also in an all-solid-state cell. $50SnO \cdot 50B_2O_3$ glass showed two plateaus around 1.5 and 0.5 V (*vs.* Li^+/Li) on the first charge (lithium insertion) and exhibited a high charge capacity of 1240 mAh g^{-1} in the liquid electrolyte system.

To clarify the lithium insertion mechanism, the local structure of the electrochemically lithium-inserted $50SnO \cdot 50B_2O_3$ glass was analyzed by XRD, ^{11}B and ^{7}Li MAS-NMR, and ^{119}Sn Mössbauer measurements. On the first plateau (1.5 V *vs.* Li^+/Li), metallic Sn with small domains was formed during the lithium insertion process. The coordination states of boron in the glass network were not changed in this stage. On the second plateau (0.5 V *vs.* Li^+/Li), Li-Sn alloy phases were formed with an increase in the inserted lithium content. The borate glass network was rearranged by the transformation from tetrahedral to trianglar boron units, which provides an additional free space compensating the volume increase of Li-Sn alloy domains.

The charge-discharge capacities of $SnO\text{-}B_2O_3$ glasses were closely related to those glass structures. The capacities were maximized at the composition with 50 mol % SnO, where the fraction of tetrahedral boron units was also maximized. This is because the glasses with large amounts of tetrahedral units could effectively utilize Sn active sites.

The all-solid-state cell consisting of $50SnO \cdot 50B_2O_3$ glass (anode), $95(0.6Li_2S \cdot 0.4SiS_2) \cdot 5Li_3BO_3$ glass (electrolyte), and $LiCoO_2$ (cathode) worked as a rechargeable battery. The $SnO\text{-}B_2O_3$ glasses are suitable anode materials with high capacity for solid-state lithium rechargeable batteries.

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